

Total No. of Questions : 8]

SEAT No. :

PA-1513

[Total No. of Pages : 2

[5926]-133

T.E. Information Technology

SOFTWARE MODELING AND DESIGN

(2019 Pattern) (Semester - II) (314454D)(Elective-II)

Time : 2½ Hours]

[Max. Marks : 70

Instructions to the candidates :

- 1) Answer Q1 or Q2,. Q3 or Q4, Q5 or Q6, Q7 or Q8.
- 2) Neat diagrams must be drawn wherever necessary.
- 3) Figures to the right side indicates full marks.
- 4) Assume Suitable data if necessary.

- Q1) a)** Draw an activity diagram for a Passport Management System To get a new passport, an applicant has to apply on-line, get the appointment. He has to submit the documents in passport office on the date of appontment. In case of insufficient or incorrect documents, the applicants have to reapply and get new appointment after submission of documents, applicants verification is done by the police. On successful verification, passport is issued to the applicant. If verification is unsuccessful, applicant has to reapply for passport. Construct the activity diagram of the given description using swim lanes. **[6]**
- b) Explain the difference between sequence & collaboration diagram with examples. **[6]**
- c) Draw a state machine diagram for an automated Washing Machine. Make suitable assumptions and clearly state the assumptions made. **[6]**

OR

- Q2) a)** The project leader schedules a meeting of members of project group by using a meeting scheduler. These are some of the assumptions:
- The project leader interacts through a GUI form to schedule the meeting. A scheduler does the automated scheduling of meeting based on the free slots in the timetable. All the members involved in the meeting will get an invitation through SMS on their mobiles. The system depends on an external mobile gateway to forward SMS. Draw a sequence diagram that describes the above system. **[9]**

P.T.O.

- b) Explain the following by drawing a neat fragment on a state diagram [9]
 - i) Entry and exit actions of a state.
 - ii) History states.
 - iii) Nested states.
 - iv) Activities.
- Q3) a)** Explain object-relation mapping and capabilities to be defined. [5]
- b) What are the activities involved in access layer design process? Explain in brief. [6]
- c) List and explain the steps in view layer macro process? [6]

OR

- Q4) a)** State and explain in detail main activities involved in design process? [9]
- b) Define OCL? What do you mean by expressions? Give the syntax for some common expressions. [8]
- Q5) a)** What is a behavioural design pattern? How iterator pattern solves the problem of traversal of any type of list? [6]
- b) Explain the solution of facade pattern using UML diagram. [6]
- c) Give the solution for structural design pattern. [6]

OR

- Q6) a)** What is an iterator pattern, what is its use and how is it modelled using UML. [9]
- b) What is a design pattern? How do you categorize design pattern? Give an example for each category. [9]
- Q7) a)** Explain the need for software architectural styles. List out some of the common architectural styles in use and give examples for each style. [6]
- b) Describe the process of messaging with SOAP and Atomic transaction. [6]
- c) Write note on component base software architecture. [5]

OR

- Q8) a)** What is software architecture and Explain software design levels. [9]
- b) Examine the following [8]
 - i) Data Abstraction and Object-Oriented Organization
 - ii) Pipes and Filters.

